Coherent Algeraic Sheaves

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Chapter 1

Sheaves

1.1 Operations on Sheaves

1.1.1 Definition of a sheaf

Let X be a topological space. A *sheaf of abelian groups* on X (or simply a *sheaf*) consist of:

(a) A *function* $x \to \mathcal{F}_x$ *sending every* $x \in X$ to an abelian group $\mathcal{F}_x$,

(b) A topoology on the set $\mathcal{F}$, the sum of the sets $\mathcal{F}_x$.

If f is an element of $\mathcal{F}_x$, we put $\pi(f) = x$; the map π is called the projection of $\mathcal{F}$ on X; the subset of $\mathcal{F} \times \mathcal{F}$ formed the pairs (f, g) such that $\pi(f) = \pi(g)$ is denoted $\mathcal{F} + \mathcal{F}$.

With these definitions in place, we can state the two axioms to which the data (a) and (b) are subject:

(I) For all $f \in \mathcal{F}$, there exist open neighborhoods V of f and U of $\pi(f)$ such that the restriction of π at V is a homeomorphism of V onto U. (In other words, if *pi* is a local homeomorphism)

(II) The mapping $f \to -f$ is a continuous mapping of $\mathcal{F}$ into $\mathcal{F}$, and the mapping $(f, g) \to f + g$ is a continuous mapping of $\mathcal{F} + \mathcal{F}$ into $\mathcal{F}$.

It will be observed that, even if X is separated (which we do not assume), $\mathcal{F}$ is not necessarily separated, which is illustrated by the example of germs of functions.

Example of a sheaf. For an abelian group G, let us set $\mathcal{F}_x = G$ for every $x \in X$; the set $\mathcal{F}$ can be identified with the product $X \times G$, and, if we equip is with the product topology of X by the discrete topology of G, we obtain a sheaf, called the *constant sheaf* isomophic with G, often identified with G.

1.1.2 Sections of a sheaf